

3 Quantum dynamics of molecules

3.1 Born-Oppenheimer approximation

The Born-Oppenheimer approximation has been introduced in previous lectures. Briefly, we write the full molecular Hamiltonian in Cartesian coordinates:

$$\begin{aligned}\hat{H} &= \sum_i -\frac{\nabla_{e,i}^2}{2} + \sum_i -\frac{\nabla_{N,i}^2}{2M_i} + \left(\sum_{j>i} \frac{1}{|\vec{r}_i - \vec{r}_j|} + \sum_{j>i} \frac{Z_i Z_j}{|\vec{R}_i - \vec{R}_j|} - \sum_{i,j} \frac{Z_j}{|\vec{r}_i - \vec{R}_j|} \right) \\ &\equiv \hat{T}_e + \hat{T}_N + \hat{V}_e + \hat{V}_N + \hat{V}_{eN}\end{aligned}\quad (3.1)$$

in atomic units.

- $\{\vec{r}, \nabla_e\}$ refers to electrons.
- $\{\vec{R}, \nabla_N\}$ refers to nuclei with masses M_i .

The TISE in the full coordinate space of nuclei and electrons is

$$\hat{H}(\vec{r}, \vec{R})\psi(\vec{r}, \vec{R}) = E\psi(\vec{r}, \vec{R}). \quad (3.2)$$

Before tackling the solution of this equation, let's consider a simpler problem of solving just the electronic part with nuclei at a fixed configuration $\vec{R}$.

$$\hat{H}_e\varphi(\vec{r}; \vec{R}) = V(\vec{R})\varphi(\vec{r}; \vec{R}), \quad (3.3)$$

where $\hat{H}_e = \hat{T}_e + \hat{V}_e + \hat{V}_{eN} + \hat{V}_N$.

We then assume that the full molecular wave function (adiabatic wave function) can be written as a product of this electronic wave function and a nuclear wave function:

$$\psi(\vec{r}, \vec{R}) = \varphi(\vec{r}; \vec{R})\chi(\vec{R}). \quad (3.4)$$

The reasoning for this assumption is that the electron mass is much smaller than the masses of the nuclei and thus that they should relax on a fast time-scale such that the nuclear dynamics effectively only feel the electrons in a mean-field sense. However, as we shall see this argument sometimes catastrophically breaks down.

Substituting this ansatz into full TISE equation (3.2) we obtain

$$\begin{aligned}\hat{H}(\vec{r}, \vec{R})[\varphi(\vec{r}; \vec{R})\chi(\vec{R})] &= V(\vec{R})\varphi(\vec{r}; \vec{R})\chi(\vec{R}) \\ &\quad - \sum_i \frac{1}{2M_i} \left[\varphi(\vec{r}; \vec{R})\nabla_i^2\chi(\vec{R}) + 2\nabla_i\varphi(\vec{r}; \vec{R})\nabla_i\chi(\vec{R}) + \nabla_i^2\varphi(\vec{r}; \vec{R})\chi(\vec{R}) \right] \quad (3.5)\end{aligned}$$

$$= E\varphi(\vec{r}; \vec{R})\chi(\vec{R}), \quad (3.6)$$

where ∇_i is used to represent $\nabla_{N,i}$.

The last two terms are proportional to m_e/M_N to some power and are therefore several orders of magnitude smaller than the other terms. Neglecting them yields

$$\hat{H}_{\text{B.O.}}(\vec{R})\chi(\vec{R}) = [\hat{T}_N + V(\vec{R})]\chi(\vec{R}) = E\chi(\vec{R}), \quad (3.7)$$

which is a TISE for nuclei under an adiabatic Hamiltonian whose effective potential is $V(\vec{R})$; this is the potential-energy surface. The set of approximations leading to equation (3.7) are known as the Born-Oppenheimer approximation.

Often classical mechanics is used to approximate the nuclear dynamics, and one simply uses the classical version of the Born-Oppenheimer Hamiltonian, $\sum_i P_i^2/2M_i + V(\vec{R})$.

3.2 Harmonic approximation

If the potential-energy surface exhibits a minimum at $\vec{R}_0$, and the dynamics is known to oscillate with small amplitudes around it, it is a commonly used approximation to replace the PES by a harmonic oscillator. This is typically done by truncating the Taylor series at second order

$$V(\vec{R}) = V(\vec{R}_0) + \frac{1}{2}(\vec{R} - \vec{R}_0)^T \cdot \nabla^2 V(\vec{R}_0) \cdot (\vec{R} - \vec{R}_0) + \dots, \quad (3.8)$$

where there is no first order term as $\nabla V(\vec{R}_0) = 0$ by definition. The Hamiltonian is therefore

$$H(\vec{P}, \vec{R}) = \sum_i \frac{P_i^2}{2M_i} + V_0 + \frac{1}{2}(\vec{R} - \vec{R}_0)^T \mathbf{K} (\vec{R} - \vec{R}_0), \quad (3.9)$$

where $V_0 = V(\vec{R}_0)$ and the Hessian is $\mathbf{K} = \nabla^2 V(\vec{R}_0)$.

Next we take the normal-mode transform

$$\vec{p} = \mathbf{U}^T \mathbf{M}^{-1/2} \vec{P} \quad (3.10)$$

$$\vec{q} = \mathbf{U}^T \mathbf{M}^{1/2} (\vec{R} - \vec{R}_0), \quad (3.11)$$

where $\mathbf{M}$ is a diagonal matrix with elements M_i along the diagonal and the orthogonal matrix $\mathbf{U}$ has columns which are the eigenvectors of the mass-weighted Hessian such that $\mathbf{U}^T \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2} \mathbf{U}$ is diagonal with values λ_k along the diagonal.

One thus has a set of independent harmonic oscillators:

$$H(\vec{q}, \vec{p}) = V_0 + \sum_k \frac{1}{2}(p_k^2 + \lambda_k q_k^2). \quad (3.12)$$

3.3 Adiabatic and diabatic representation

The description of molecular dynamics in electronically excited states almost systematically requires a description beyond the Born-Oppenheimer approximation. Such descriptions can be made either in the (unique) adiabatic representation or in one (of infinitely many) diabatic representations. The ensemble of electronic states of any of these representations forms a complete basis.

3.3.1 Adiabatic representation

It is possible to go beyond the Born-Oppenheimer approximation by writing a formally exact solution*

$$\psi(\vec{r}, \vec{R}) = \sum_n \varphi_n(\vec{r}; \vec{R}) \chi_n(\vec{R}). \quad (3.13)$$

*we neglect here the continuum which is, in principle, required

where $\varphi_n(\vec{r}; \vec{R})$ are the set of all electronic states solving

$$\hat{H}_e \varphi_n(\vec{r}; \vec{R}) = V_n(\vec{R}) \varphi_n(\vec{r}; \vec{R}) \quad (3.14)$$

Substituting this ansatz equation (3.13) into equation (3.2), multiplying from the left with $\varphi_m(\vec{r}; \vec{R})$ and integrating over the electronic coordinates yields a set of coupled equations:

$$\sum_n \langle \varphi_m | \hat{H} | \varphi_n \rangle \chi_n(\vec{R}) = \sum_n \langle \varphi_m | \hat{T}_N | \varphi_n \rangle \chi_n(\vec{R}) + \sum_n \langle \varphi_m | \varphi_n \rangle V_n(\vec{R}) \chi_n(\vec{R}) \quad (3.15)$$

$$= [\hat{T}_N + V_m(\vec{R})] \chi_m(\vec{R}) - \sum_n \hat{\Lambda}_{mn} \chi_n(\vec{R}) \quad (3.16)$$

$$= E \chi_m(\vec{R}) \quad (3.17)$$

where $\hat{\Lambda}_{mn}$ are the nonadiabatic couplings defined as

$$\hat{\Lambda}_{mn} = \delta_{mn} \hat{T}_N - \langle \varphi_m | \hat{T}_N | \varphi_n \rangle. \quad (3.18)$$

For example, taking only the lowest two levels, the equation can be written in matrix notation as

$$\begin{pmatrix} \hat{T}_N + V_1(\vec{R}) - \hat{\Lambda}_{00} & -\hat{\Lambda}_{01} \\ -\hat{\Lambda}_{10} & \hat{T}_N + V_2(\vec{R}) - \hat{\Lambda}_{11} \end{pmatrix} \begin{pmatrix} \chi_1(\vec{R}) \\ \chi_2(\vec{R}) \end{pmatrix} = E \begin{pmatrix} \chi_1(\vec{R}) \\ \chi_2(\vec{R}) \end{pmatrix} \quad (3.19)$$

Inserting the position-representation of the nuclear kinetic energy operator gives

$$\hat{\Lambda}_{mn} = \frac{1}{2M} [2\hat{F}_{mn} \cdot \nabla + \hat{G}_{mn}] \quad (3.20)$$

$$\hat{F}_{mn} = \langle \varphi_m | \nabla \varphi_n \rangle \quad (3.21)$$

$$\hat{G}_{mn} = \langle \varphi_m | \nabla^2 \varphi_n \rangle \quad (3.22)$$

with Dirac brackets referring only to integration over electronic coordinates. Note that these are operators as they include ∇ , which is a symbol for the multi-dimensional derivative with respect to mass-weighted nuclear coordinates, and to ease the notation, we have assumed that the coordinates have been mass-weighted such that M is the same for all degrees of freedom. $\hat{F}_{mn}$ is a vector with a number of components equal to the number of nuclear degrees of freedom; $\hat{G}_{mn}$ is a scalar. Since the electronic wave functions are normalized at all nuclear geometries, $\langle \varphi_m | \varphi_n \rangle = \delta_{mn}$, it follows that $\hat{F}_{mn}$ is a anti-Hermitian matrix, *i.e.*

$$\hat{F}_{mn} = -\hat{F}_{nm}^*. \quad (3.23)$$

Since electronic wave functions can always be chosen real-valued, it means that diagonal effects of $\hat{F}_{mn}$ vanish: $\hat{F}_{nn}(\vec{R}) = 0$.

As a first approximation, we could keep only the diagonal terms in equation (3.15). We would then obtain the Born–Oppenheimer Schrödinger equation for the nuclei with an additional contribution from $\hat{G}_{nn}$ which is usually small but non-vanishing

$$\left[\hat{T}_N + V_n(\vec{R}) - \frac{\hat{G}_{nn}(\vec{R})}{2M} \right] \chi_n(\vec{R}) = E \chi_n(\vec{R}) \quad (3.24)$$

This is known as the Born-Huang approximation.

In order to find out if we can do this, we should consider the magnitude of the off-diagonal components. Thinking of the nonadiabatic coupling operators as matrices, one can show that $\mathbf{G} = (\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot \mathbf{F}$ and therefore $\hat{T}_N - \Lambda = -\frac{1}{2M}(\nabla + \mathbf{F})^2$. Using this, one does not have to worry about the nonadiabatic scalar couplings G_{nm} anymore.

The question is therefore whether $\hat{F}_{mn}$ is large or small. To find out, we apply first ∇ and then $\langle \varphi_m |$ (where $m \neq n$) to the electronic Schrödinger equation

$$\nabla H_e |\varphi_n\rangle = \nabla V_n |\varphi_n\rangle \quad (3.25)$$

$$(\nabla H_e) |\varphi_n\rangle + H_e |\nabla \varphi_n\rangle = (\nabla V_n) |\varphi_n\rangle + V_n |\nabla \varphi_n\rangle \quad (3.26)$$

$$\langle \varphi_m | (\nabla H_e) |\varphi_n\rangle + \langle \varphi_m | H_e |\nabla \varphi_n\rangle = (\nabla V_n) \langle \varphi_m | \varphi_n\rangle + V_n \langle \varphi_m | \nabla \varphi_n\rangle \quad (3.27)$$

$$\langle \varphi_m | (\nabla H_e) |\varphi_n\rangle + V_m \langle \varphi_m | \nabla \varphi_n\rangle = (\nabla V_n) \times 0 + V_n \langle \varphi_m | \nabla \varphi_n\rangle \quad (3.28)$$

and therefore

$$\hat{F}_{mn} = \frac{\langle \varphi_m | (\nabla H_e) |\varphi_n\rangle}{V_n - V_m}. \quad (3.29)$$

This will become large whenever the states n and m become close in energy, e.g. at avoided crossings. Near these configurations, the Born-Oppenheimer approximation breaks down and it may be possible for the wave function to flow from one state into another. At conical intersections where $V_n = V_m$, the expression blows up. It is therefore necessary to find a different representation which is more convenient.

3.3.2 Diabatic representation

The alternative to the adiabatic basis is a diabatic basis which is chosen such that all electronic matrix elements of the derivatives $\frac{\partial}{\partial \vec{R}}$ and $\frac{\partial^2}{\partial \vec{R}^2}$ are as small as possible. There are infinitely many ways of choosing a diabatic basis. The simplest version consists in choosing the electronic basis functions to be independent of the nuclear geometry $\varphi(\vec{r}; \vec{R}) \rightarrow \varphi(\vec{r}; \vec{R}_0)$. This is called the “crude adiabatic” basis in which

$$\psi(\vec{r}, \vec{R}) = \sum_n \varphi_n(\vec{r}; \vec{R}_0) \chi_n^{(0)}(\vec{R}). \quad (3.30)$$

The basis functions $\varphi_n(\vec{r}; \vec{R}_0)$ are solutions of the electronic Schrödinger equation at geometry $\vec{R}_0$:

$$\hat{H}_e(\vec{r}; \vec{R}_0) \varphi_n(\vec{r}; \vec{R}_0) = [\hat{T}_e + V(\vec{r}; \vec{R}_0)] \varphi_n(\vec{r}; \vec{R}_0) \quad (3.31)$$

$$= V_n(\vec{R}_0) \varphi_n(\vec{r}; \vec{R}_0) \quad (3.32)$$

where $V(\vec{r}; \vec{R}_0)$ includes $\hat{V}_e$, $\hat{V}_N$, and $\hat{V}_{eN}$. Substituting equation (3.30) into equation (3.2), multiplying from the left with $\varphi_m(\vec{r}; \vec{R}_0)$ and integrating over the electronic degrees of freedom we obtain

$$\hat{H} \psi(\vec{r}, \vec{R}) = [\hat{T}_N + \hat{H}_e(\vec{r}; \vec{R})] \psi(\vec{r}, \vec{R}) \quad (3.33)$$

$$= [\hat{T}_N + \hat{H}_e(\vec{r}; \vec{R}_0) + V(\vec{r}; \vec{R}) - V(\vec{r}; \vec{R}_0)] \psi(\vec{r}, \vec{R}) \quad (3.34)$$

$$\langle \varphi_m | \hat{H} | \psi \rangle = \sum_n \langle \varphi_m | \hat{T}_N + V_n(\vec{R}_0) + V(\vec{r}; \vec{R}) - V(\vec{r}; \vec{R}_0) | \varphi_n \rangle \chi_n^{(0)}(\vec{R}) \quad (3.35)$$

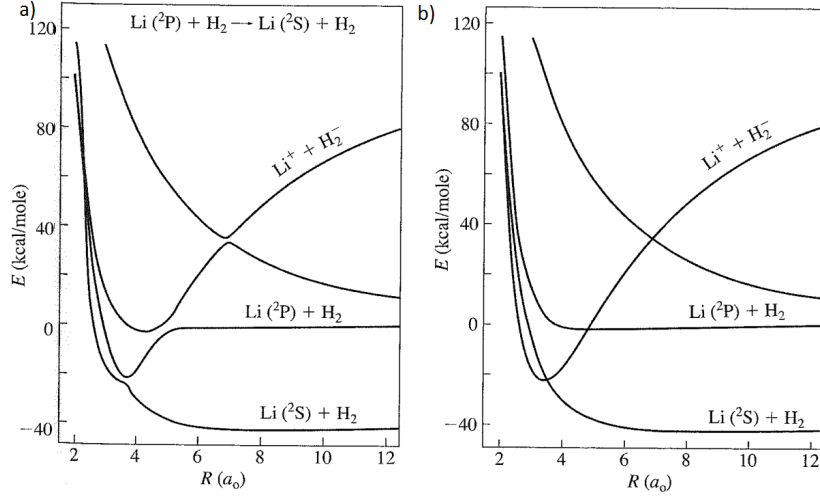


Figure 2: Slices of the four lowest potential energy surfaces of a particular symmetry of LiH_2 as a function of the Li- H_2 separation. The orientation of the H_2 internuclear axis is 60° relative to the Li- H_2 direction, and the H_2 internuclear distance is $1.4 \text{ \AA}$ (adiabatic representation). Note that the adiabatic potentials, being of the same symmetry, obey a “non-crossing” rule. At the avoided crossings, the physical characters of the adiabatic electronic states generally change, for example, from ionic to covalent. (b) Diabatic representation, potentials have continuous physical character. (taken from Introduction to quantum mechanics, David J. Tannor)

and therefore

$$\sum_n \{\hat{T}_N \delta_{nm} + U_{mn}(\vec{R})\} \chi_n^{(0)} = E \chi_m^{(0)}, \quad (3.36)$$

where $U_{mn}(\vec{R}) = \langle \varphi_m(\vec{R}_0) | V_n(\vec{R}_0) + \hat{V}(\vec{r}, \vec{R}) - \hat{V}(\vec{r}, \vec{R}_0) | \varphi_n(\vec{R}_0) \rangle$ is the coupling between the states. As $\hat{H}$ must be Hermitian, so is the diabatic coupling matrix, $U_{mn} = U_{nm}^*$. (Again, brackets imply integration over the electronic coordinates only).

3.3.3 Converting between representations

The adiabatic representation is complicated due to the $\hat{\Lambda}_{nm}$ operators which involve derivatives. In certain cases it is possible to transform the states into a diabatic representation where the derivative couplings disappear. This is not always possible, but sometimes the transformation makes them small enough that they can be neglected.

Let us consider the simplest case where there are only two relevant adiabatic states and suggest the transformation

$$\begin{pmatrix} \tilde{\varphi}_1(\vec{r}; \vec{R}) \\ \tilde{\varphi}_2(\vec{r}; \vec{R}) \end{pmatrix} = \mathbf{S}(\vec{R}) \begin{pmatrix} \varphi_1(\vec{r}; \vec{R}) \\ \varphi_2(\vec{r}; \vec{R}) \end{pmatrix} \quad \mathbf{S}(\vec{R}) = \begin{pmatrix} \cos \gamma(\vec{R}) & \sin \gamma(\vec{R}) \\ -\sin \gamma(\vec{R}) & \cos \gamma(\vec{R}) \end{pmatrix} \quad (3.37)$$

where we will choose function $\gamma(\vec{R})$ later.

Consider the $\hat{F}$ operator in the new basis:

$$\tilde{F}_{12}(\vec{R}) = \langle \tilde{\varphi}_1 | \nabla \tilde{\varphi}_2 \rangle \quad (3.38)$$

$$= \sum_{nm} \langle \varphi_n | S_{1n} (\nabla S_{2m}) | \varphi_m \rangle + \sum_{nm} \langle \varphi_n | S_{1n} S_{2m} | \nabla \varphi_m \rangle \quad (3.39)$$

$$= -\nabla \gamma(\vec{R}) + F_{12}(\vec{R}), \quad (3.40)$$

where the final expression follows from differentiating the transformation matrix and using trigonometric identities.

If we choose $\gamma(\vec{R})$ such that equation (3.40) is equal to 0, then we will have removed the derivative coupling terms. This is easy to do for a one-dimensional system by choosing $\gamma(R) = \int_0^R F_{12}(R') dR'$.

Therefore in the diabatic basis, our Schrödinger equation becomes

$$\begin{pmatrix} \hat{T}_N + U_{11}(\vec{R}) & U_{12}(\vec{R}) \\ U_{21}(\vec{R}) & \hat{T}_N + U_{22}(\vec{R}) \end{pmatrix} \begin{pmatrix} \tilde{\chi}_1(\vec{R}) \\ \tilde{\chi}_2(\vec{R}) \end{pmatrix} = E \begin{pmatrix} \tilde{\chi}_1(\vec{R}) \\ \tilde{\chi}_2(\vec{R}) \end{pmatrix} \quad (3.41)$$

with the diabatic potential-energy matrix

$$\mathbf{U}(\vec{R}) = \mathbf{S}(\vec{R})^T \begin{pmatrix} V_1(\vec{R}) & 0 \\ 0 & V_2(\vec{R}) \end{pmatrix} \mathbf{S}(\vec{R}) \quad (3.42)$$

Note the simplicity of equation (3.41) compared to equation (3.19). The transformation from the adiabatic into the diabatic basis has also transformed derivative couplings $(\frac{d}{dR})_{mn}$ into coordinate couplings $(\hat{U}(R))_{mn}$. Diagonal elements of U are diabatic surfaces and off-diagonal elements are non-adiabatic couplings which can cause non-adiabatic transitions. In practice one looks for a conventional “diabatization scheme” that minimizes the coupling. The complete elimination of couplings is only possible for **diatomics**. The situation for **polyatomics** is generally more complicated and some nonadiabatic couplings remain.

Converting from the diabatic representation to the adiabatic representation is however much easier. The $\mathbf{S}$ matrix is defined uniquely as the one which diagonalizes the $\mathbf{U}$ matrix, i.e. rearrange equation (3.42).

3.4 Adiabatic dynamics: Quantum adiabatic theorem

We consider a quantum system with discrete level structure and a Hamiltonian given by:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t). \quad (3.43)$$

The matrix representation of $\hat{H}$ in the basis of eigenstates of $\hat{H}_0$ is:

$$H(t) = \begin{pmatrix} E_a & V_{ab}(t) & \dots \\ V_{ba}(t) & E_b & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

We seek a solution of the corresponding TDSE, which can be written as

$$\psi(t) = \begin{pmatrix} \psi_a(t) \\ \psi_b(t) \\ \vdots \end{pmatrix}.$$

One can define a unitary transformation that diagonalizes the instantaneous Hamiltonian $H(t)$:

$$\begin{aligned} S^\dagger(t)H(t)S(t) &= D(t) \\ S^\dagger(t)\psi(t) &= \tilde{\psi}(t), \end{aligned} \quad (3.44)$$

where $D(t)$ is a diagonal matrix and $S^\dagger(t) = S^{-1}(t)$.

The TDSE can be written as

$$i\hbar \frac{\partial}{\partial t} (S(t)\tilde{\psi}(t)) = i\hbar \left(S(t) \frac{\partial \tilde{\psi}(t)}{\partial t} + \frac{\partial S(t)}{\partial t} \tilde{\psi}(t) \right) = H(t)S(t)\tilde{\psi}(t). \quad (3.45)$$

Pre-multiplying both sides by $S^\dagger(t)$ and rearranging gives

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}(t) = D(t)\tilde{\psi}(t) - i\hbar S^\dagger(t) \frac{\partial S(t)}{\partial t} \tilde{\psi}(t). \quad (3.46)$$

N.B. $S^\dagger(t)S(t) = I$ so $0 = \frac{\partial}{\partial t} S^\dagger(t)S(t) = \frac{\partial S^\dagger(t)}{\partial t} S(t) + S^\dagger(t) \frac{\partial S(t)}{\partial t}$ so the last line can be alternatively written as

$$i\hbar \frac{\partial}{\partial t} \tilde{\psi}(t) = D(t)\tilde{\psi}(t) + i\hbar \frac{\partial S^\dagger(t)}{\partial t} S(t)\tilde{\psi}(t).$$

If $H(t)$ is slowly varying, so will be $S(t)$, and in this case $\frac{\partial S(t)}{\partial t}$ may be neglected. This is called the *adiabatic approximation*. The Born–Oppenheimer approximation is an example of an adiabatic approximation, although in this case, the off-diagonal couplings are not time dependent.

Adiabatic theorem: If $\psi(0)$ is an eigenfunction of $H(0)$ and $H(t)$ is a slowly-varying function of time, then $\psi(t)$ will evolve in such a way as to remain an eigenfunction of $H(t)$ for all time, in the limit of infinitely-slowly-varying $H(t)$. (see A.Messiah Quantum Mechanics, North Holland Publishing Company, 1967 for the detailed derivation).

3.5 Intersections between adiabatic potential-energy surfaces

Particularly interesting is the case of **conical intersections**. We examine the intersection of adiabatic surfaces. In diatomics, adiabatic curves belonging to electronic states of the same symmetry generally do not cross but exhibit avoided crossings. The proof of this non-crossing rule is simple in a diabatic basis:

$$\mathbf{U} = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}.$$

where $U_{21} = U_{12}^*$ so that the matrix is Hermitian. The off-diagonal elements are zero, unless the states 1 and 2 have the same electronic symmetry. Diagonalization gives the adiabatic potentials

$$V_{\pm} = \frac{U_{11} + U_{22}}{2} \pm \frac{1}{2} \sqrt{(U_{11} - U_{22})^2 + 4|U_{12}|^2}. \quad (3.47)$$

The energy gap between the curves is

$$\Delta V = V_+ - V_- = \sqrt{(U_{11} - U_{22})^2 + 4|U_{12}|^2} \quad (3.48)$$

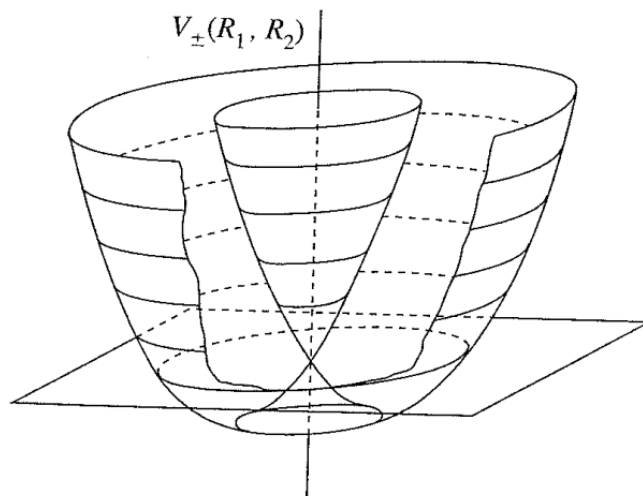


Figure 3: Potential energy functions $V_{\pm}$ as a function of two non-symmetric nuclear coordinates showing the geometry of a conical intersection. (taken from Introduction to quantum mechanics, David J. Tannor)

Curves only cross if $U_{11}(R) = U_{22}(R)$ and $U_{12}(R) = 0$ which represent two conditions to be met for one variable R . There is in general no reason for both these conditions to be met simultaneously in a one-dimensional system, leading to the non-crossing rule for diatomics. In two dimensions there is generally a single (or a discrete set of) points where the two surfaces touch leading to so-called conical intersections. In $n > 2$ dimensions, the intersection space is normally a manifold of dimension $n-2$ resulting from the two constraints given above e.g. triatomics with $3N - 6 = 3$ degrees of freedom have a one-dimensional “seam” of conical intersections.

Consider the simple case of

$$\mathbf{U} = \begin{pmatrix} ax & by \\ by & -ax \end{pmatrix} \quad (3.49)$$

and convert to polar coordinates $ax = r \cos \theta$ and $by = r \sin \theta$:

$$\mathbf{U} = \begin{pmatrix} r \cos \theta & r \sin \theta \\ r \sin \theta & -r \cos \theta \end{pmatrix}. \quad (3.50)$$

This diabatic potential-energy matrix has eigenvalues $\pm r$ and the lower (Born–Oppenheimer) state has the eigenvector $(c_1, c_2) = (\sin \frac{\theta}{2}, -\cos \frac{\theta}{2})$. The total electronic wavefunction, $\psi = c_1 \phi_1 + c_2 \phi_2$, thus changes sign when θ goes from 0 to 2π . Therefore in order for the total molecular wavefunction to be single valued everywhere, the nuclear wavefunction must also change sign. In this way, the existence of a conical intersection can have strong effects on spectroscopy and dynamics even when the dynamics occurs at energies far below the intersection and otherwise proceeds purely on the ground-state Born–Oppenheimer PES. This known as the *geometric-phase effect*.

3.6 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Adiabatic and diabatic representations

- W. Domcke and D. R. Yarkony and H. Köppel, Conical Intersections: Electronic Structure, Dynamics and Spectroscopy [chapter 1 has a good description of nonadiabatic coupling]
- Tannor, Introduction to quantum mechanics: a time-dependent perspective [chapter 12]
- May and Kühn, Charge and Energy Transfer Dynamics in Molecular Systems (Wiley, 2011)

Geometric-phase effect

- Baer, Beyond Born–Oppenheimer (Wiley, 2006) [goes deep into maths, well beyond this course, but maybe you’ll find it interesting in the future]